

SEAI — Symbolic and Evolutionary Artificial Intelligence

Partial Graph Matching for Indoor Robot Localization

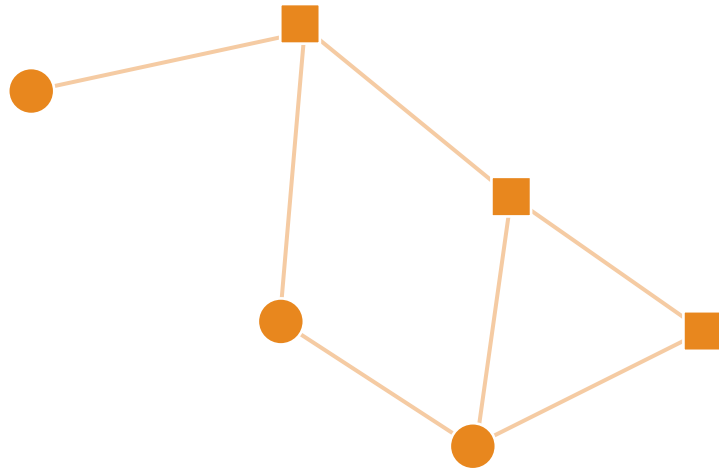
A Graph Neural Network Approach with Sinkhorn–Hungarian Assignment

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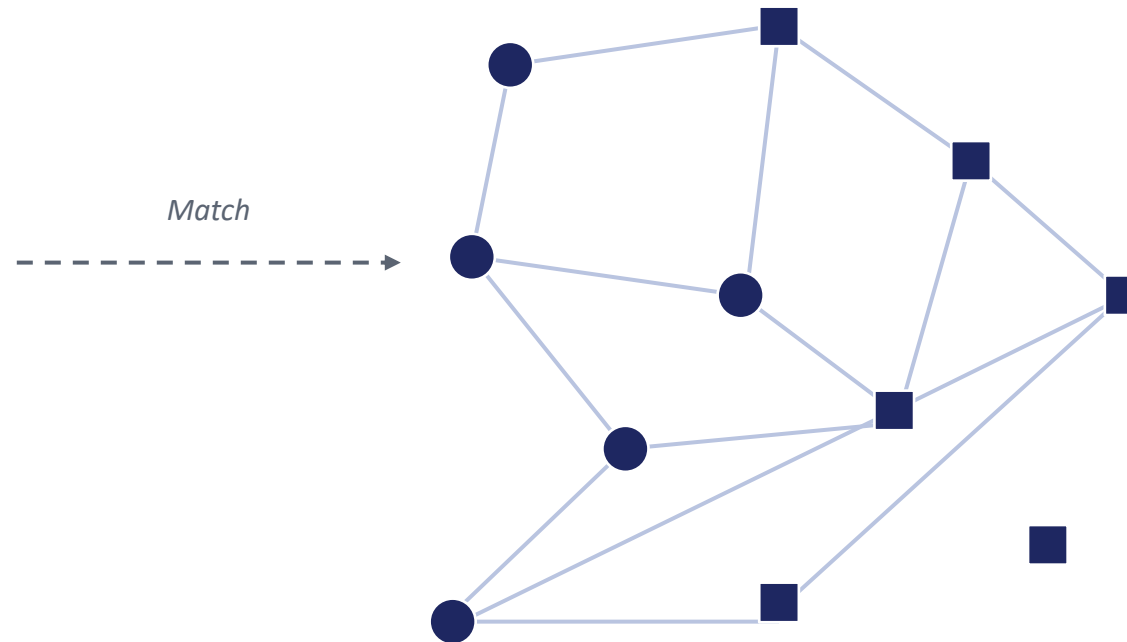
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The Problem: Matching a Partial Observation Against a Complete Model

S-Graph — robot's partial, noisy observation



A-Graph — complete architectural graph from BIM



Match



● Room node

■ Wall-surface node

Partiality

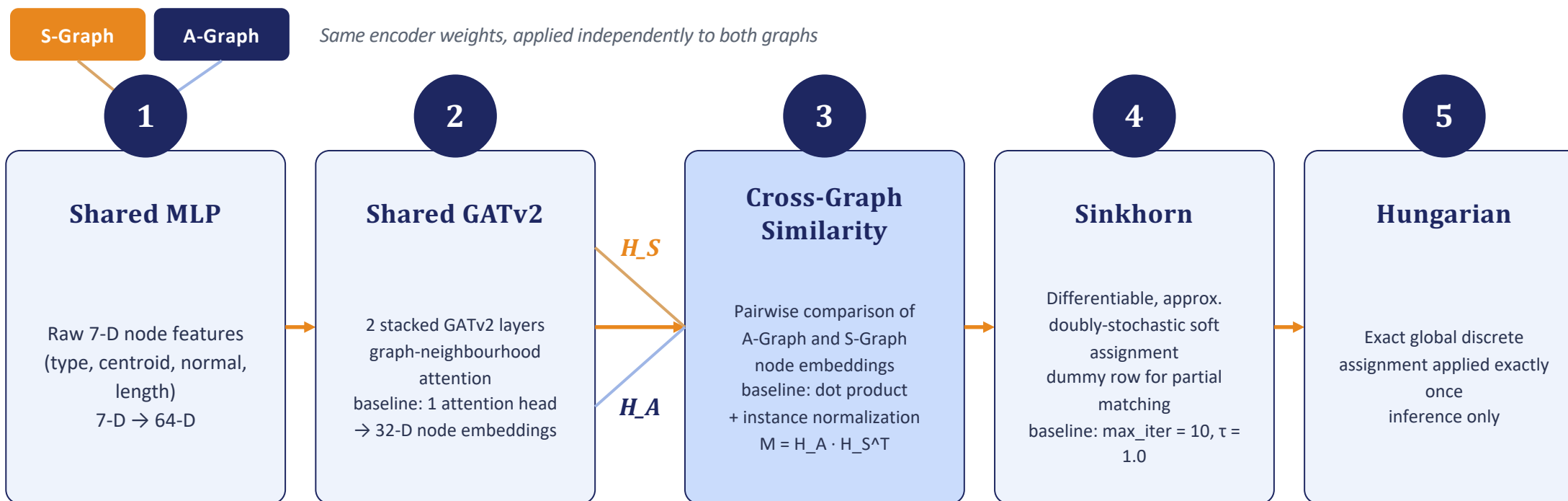
S-Graph is smaller, incomplete, and noisy — not a bijection with the A-Graph

Node heterogeneity

Rooms and wall-surfaces carry very different amounts of geometric information

The Baseline Matching Pipeline

Shared node encoder → cross-graph similarity → soft assignment → discrete matching



Single discretization: Sinkhorn remains soft; Hungarian is applied exactly once.

Methodology: Why F_1 , Not Accuracy

Choosing the right metric was itself a key finding of this project

Baseline Accuracy

≈ 0.994

saturates near 99% — regardless of matching quality

Baseline F_1

0.73–0.76

discriminates between weak and strong matching models

Accuracy is still reported as requested, but F_1 is used for model comparison.

Why the gap: extreme class imbalance

- Each S-Graph node has ≈ 88 candidate A-Graph nodes on average, but only one correct match.
- ≈ 6.65M of 6.75M test-set cells are negatives.
- Accuracy is dominated by true negatives; F_1 focuses on the positive matching decisions.

Beyond F_1 : Supporting Metrics and Evaluation Views

Not every metric survived — each one had to prove it could actually distinguish a real result

Sinkhorn vs. Hungarian — soft vs. discrete evaluation

Sinkhorn output: differentiable soft assignment matrix; evaluated before final discretization.

Hungarian output: global discrete assignment enforcing an exact one-to-one matching.

We report both throughout: Sinkhorn metrics show the quality of the soft learned assignment, while Hungarian metrics reflect the model's final operating point.

Two supporting metrics, one real cross-check

mean_errors_per_graph

Average matching errors per graph.

Stable supporting metric; usually agreed with F_1 .

perfect_pct / n_perfect_graphs

Fraction/count of perfectly matched graphs.

Useful cross-check, but noisier because each graph contributes only a binary outcome.

Supporting metrics are used as independent cross-checks, not as decision criteria on their own.

Not used for model comparison: Precision/Recall are mathematically redundant with F_1 in this matching structure; per-type accuracy suffers from the same saturation as global accuracy. Global accuracy is retained as requested.

Baseline Configuration: Choosing the Scheduler

Primary metrics favor the scheduler; the perfect-graph count provides a useful dissenting signal

Metric	No scheduler	With scheduler	Winner
F ₁ (global)	0.7497	0.7565	scheduler
mean errors / graph	5.2004	5.0594	scheduler
Perfect graphs	4.17% (153)	3.92% (144)	no scheduler

Both primary metrics favor the scheduler; only the noisier perfect-graph metric slightly favors the unscheduled run.

Check 1 — the scheduled run

LR reduction fires once at epoch 180/200 ($1e-3 \rightarrow 5e-4$). The best checkpoint is the final epoch: validation loss improves from 0.021049 before the LR cut to 0.020380 after it (~3.2%).

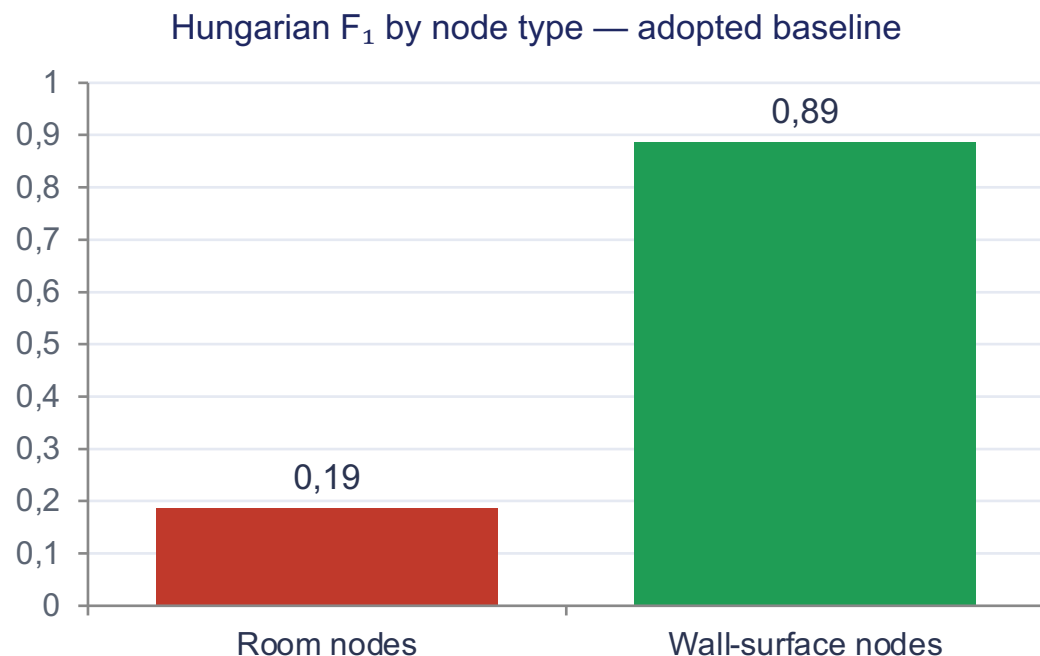
Check 2 — the unscheduled run, checked too

Without the scheduler, the best checkpoint is epoch 190, not the final epoch. Validation loss rises from the best 0.020817 to 0.021498 at epoch 199, with noisy oscillations over the final ~30 epochs.

Decision: adopt the scheduler baseline.

Baseline: A Persistent Room vs. Wall-Surface Gap

Discovered early, and central to every decision in the rest of this study



Why:

- For room nodes, the centroid is the only informative geometric feature; normal and length are placeholders
- Wall-surface nodes carry true geometric descriptors (real normal + length)
- Rooms are harder to distinguish, especially when several rooms have similar geometry or structural context

This gap shapes the rest of the study: we report results by node type, prioritize room matching, and interpret every later experiment through this asymmetry.

Experimental Cascade: One Motivated Change at a Time

Five motivated modifications, evaluated incrementally against the current reference configuration



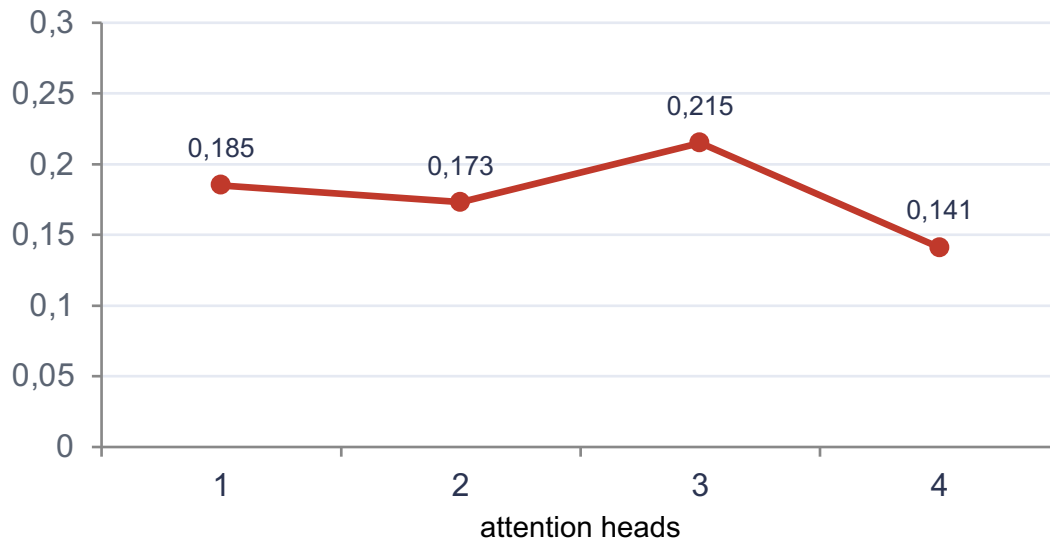
Guiding rule

Each modification was first evaluated against the current reference configuration and retained only when supported by the results. Interactions were investigated when experimental evidence motivated them.

This cascade later exposed a strong non-additive interaction and an unexpected reversal in the leave-one-out analysis.

Steps 1–2: Attention Heads, then Cosine Similarity

Room F₁ (Hungarian) — non-monotonic across attention heads



Selected: heads = 4 — strongest overall metrics (global F₁ 0.757 · WS F₁ 0.897 · perfect graphs 6.48%). Heads = 3 wins only on the more variable room F₁.

Cosine Similarity — the largest individual improvement

Room F₁ (Hungarian): dot product → cosine

0.141 → **0.526** $\approx 3.7\times$

Improves every reported metric simultaneously, with no observed trade-off.

- L2-normalizing the embeddings turns the dot product into cosine similarity, removing dependence on embedding magnitude.

Steps 3–4: Two Targeted Interventions — Both Fail Alone

One loss-level fix and one assignment-level fix — both harmful in isolation

Weighted BCE — linear

$$w = N_1 - 1$$

Reweight the single positive match per column against the many negatives.

0.526 → **0.281**

linear weight, alone — Room F₁, Hungarian

Interpretation: the positive weight reaches 309 on the largest graphs, producing an overly aggressive training signal.

Sinkhorn Iterations: 10 → 30

30 iterations selected through empirical Sinkhorn convergence analysis.

0.526 → **0.253**

max_iter = 30, alone — Room F₁, Hungarian

Interpretation: max_iter also changes the training-time gradient path; a sharper Sinkhorn output may destabilize optimization when used alone.

THE KEY FINDING

A Non-Additive Interaction: Two Failures Succeed Together

Current reference (cosine) 0.526

Linear weighted BCE alone 0.281

max_iter = 30 alone 0.253

Both together

0.742

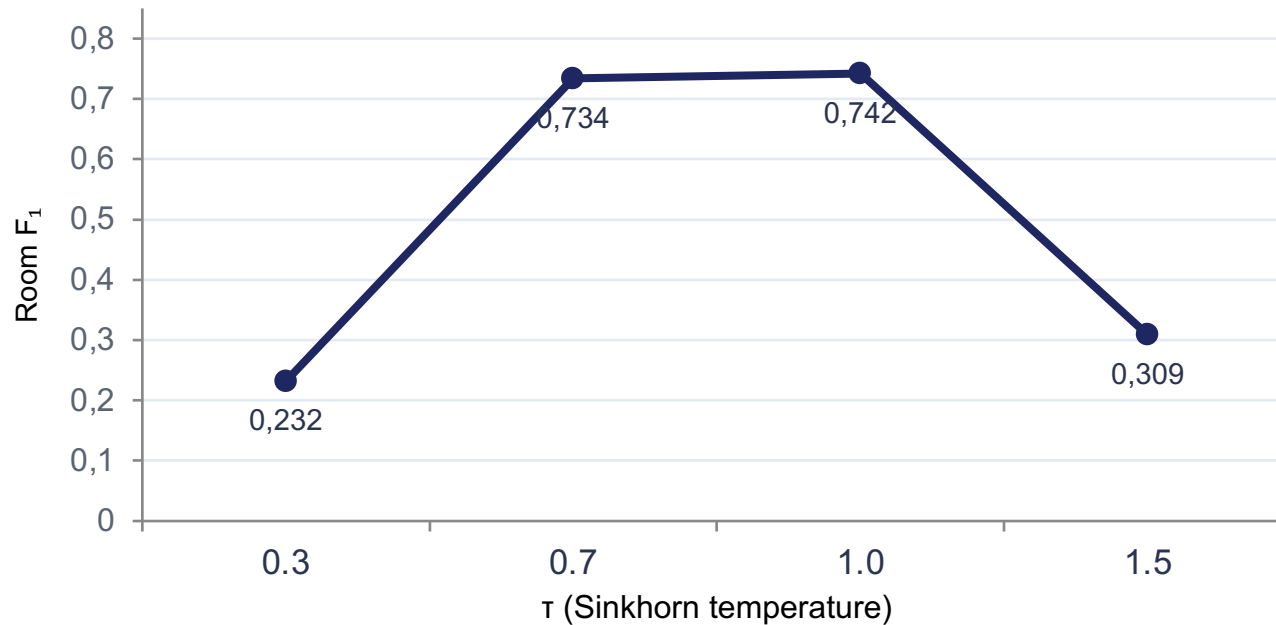
Control: sqrt weighting + max_iter = 30 $\rightarrow$ room F_1 = 0.298 — the interaction needs the aggressive linear weight specifically.

Proposed interpretation: weighted BCE becomes effective only when paired with a sufficiently sharp Sinkhorn output; max_iter = 30 provides that regime.

Step 5: Sinkhorn Temperature — A Narrow Optimum Around $\tau = 1.0$

An inference-only sweep suggested lower τ helps monotonically — full training tells a different story

Full-training room F_1 (Hungarian) vs. Sinkhorn temperature



Why the cheap sweep was misleading

Inference-only sweep (no retraining): $\tau = 0.3$ appeared best.

Full retraining: $\tau = 0.3$ collapses to room $F_1 = 0.232$.

Reading

- $\tau = 1.0$ is the best tested value after full retraining — not merely the unchanged default.
- Moving sufficiently far in either direction causes a sharp degradation.
- The original hypothesis that a softer assignment (higher τ) could help ambiguous room matches is not supported.
- Inference-only sweeps can be misleading when the parameter also affects training-time gradients — the same lesson observed with `max_iter`.

Final Cascade Configuration

Scheduler	Heads	Similarity	Loss	Sinkhorn Iterations	Sinkhorn Temperature
ReduceLROnPlateau	4	cosine	weighted BCE (linear)	max_iter = 30	$\tau = 1.0$

Overall progress from the adopted baseline

F ₁ global	F ₁ room	F ₁ wall-surface	Perfect matches
0.757	0.185	0.886	3.92%
↓	↓	↓	↓
0.901	0.742	0.937	28.02%

Leave-One-Out: Isolating Each Component's Contribution in the Final Configuration

Requested by the course supervisor — remove exactly one component from the final configuration, hold everything else fixed

Why this differs from testing components in isolation earlier

- Cascade: each modification was evaluated against the reference configuration available at that stage.
- Leave-one-out: remove one component from the final configuration while holding all others fixed, measuring its marginal contribution in the finished pipeline.

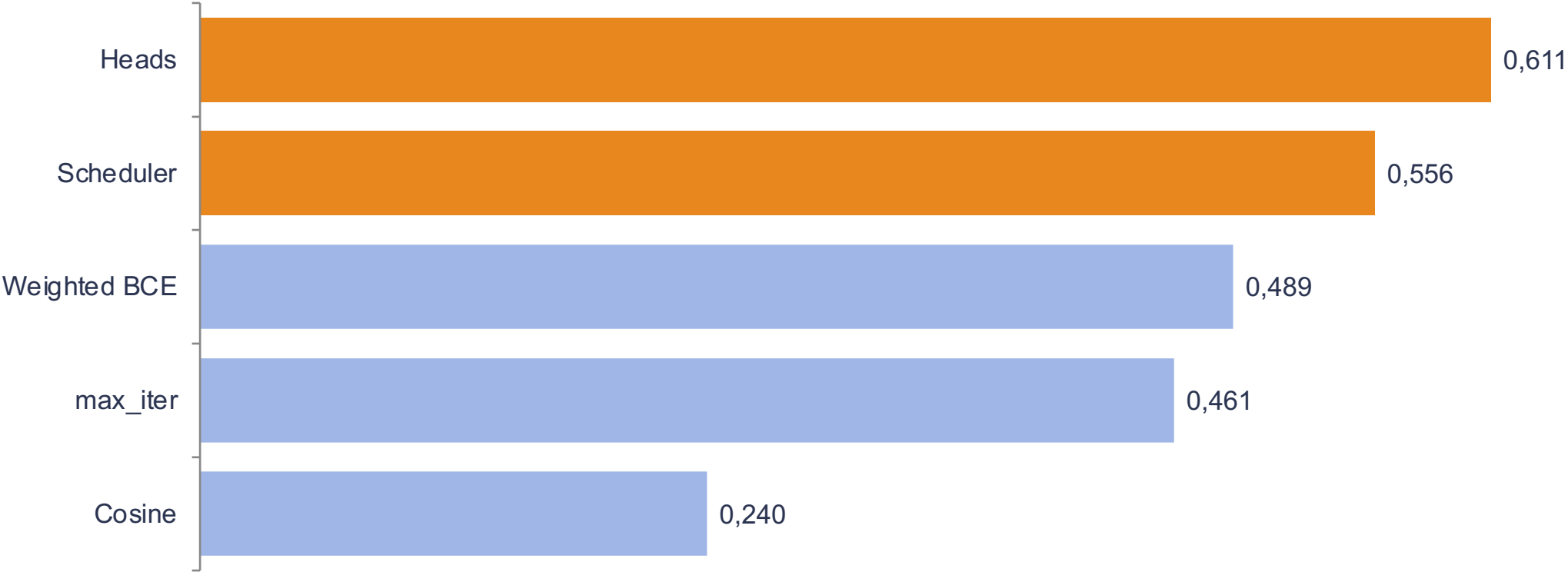
Three leave-one-out cases reuse existing results

- **Temperature τ :** the retained $\tau = 1.0$ equals the framework default — removing the intervention changes nothing. LOO contribution = 0 by construction. This does not mean temperature is irrelevant — see the sweep on the previous slide.
- **Remove max_iter:** 30 $\rightarrow$ 10 while keeping weighted BCE = the existing weighted-BCE-alone run from the cascade.
- **Remove weighted BCE:** keeping max_iter = 30 = the existing max_iter-alone run from the cascade.

Only scheduler, attention heads, and cosine similarity required new dedicated leave-one-out training runs.

Leave-One-Out Results

Decrease in Hungarian room F_1 from the final value (0.742) when one component is removed



A striking reversal: Heads and Scheduler, seemingly marginal early on, become the two most critical components for room matching in the final pipeline.

τ not shown: the retained $\tau = 1.0$ equals the framework default, so its leave-one-out contribution is zero by construction.

A SECOND REVERSAL

Components That Looked Marginal Become Critical in the Final Pipeline

Scheduler

Early in the cascade:

Small, mixed improvement in the baseline comparison.

-0.307

In the final leave-one-out:

Removing the scheduler causes the largest drop in global F_1 . Zero inference-time cost.

Attention Heads

Early in the cascade:

Strongly non-monotonic room-specific effect; no clear room-level trend.

-0.611

In the final leave-one-out:

Reducing 4 $\rightarrow$ 1 head causes the largest drop in room F_1 .

Component importance is configuration-dependent: an intervention that looks marginal against a weak baseline can become essential once the full pipeline is in place.

Cost–Benefit and Inference Time

Requested by the course supervisor: how much does each component really help — and what does it cost at inference?

Component	ΔF_1 global	ΔF_1 room	Inference cost	Cost–benefit verdict
Scheduler	+0.307	+0.556	Zero — training only	Clearly worthwhile
Attention heads	+0.133	+0.611	Real, modest (~14% measured)	Clearly worthwhile
Cosine similarity	+0.115	+0.240	Theoretically negligible	Essentially free
Weighted BCE	+0.099	+0.489	Zero — training only	No inference trade-off
Sinkhorn iterations	+0.091	+0.461	Real, modest (~16% measured)	Worthwhile

Timing percentages are indicative: measurements were collected on a shared CPU and show non-negligible runtime noise.

Large accuracy gains did not require a large inference-time penalty: two high-impact components cost exactly zero at deployment, while the remaining overheads are modest.

Limitations

1

Single-seed experimentation

All experiments use one fixed seed; under the CPU setup, training is deterministic for that seed. The central weighted-BCE + max_iter interaction has not been confirmed across multiple random seeds.

2

Inference-only sweeps can be misleading

Both max_iter and τ showed this directly: inference-only sweeps gave misleading predictions because both parameters also affect training-time gradients.

3

Residual room-matching gap

Final F_1 : 0.742 room vs. 0.937 wall-surface. The absolute gap shrank from ~ 0.70 at baseline to ~ 0.20 in the final model, but remains substantial. No tested modification removes the underlying feature asymmetry of room nodes.

Conclusions & Future Work

What we learned — and what should be tested next

CONCLUSIONS

Metric design matters

Accuracy saturation hid meaningful differences; F_1 became the primary model-selection metric.

Interactions matter

Weighted BCE and increased Sinkhorn iterations fail alone but succeed together — a clear non-additive interaction.

Component importance is configuration-dependent

Scheduler and attention heads looked marginal early, yet become critical in the final leave-one-out.

Global F_1 : 0.757 → 0.901 · Room F_1 : 0.185 → 0.742

NEXT STEPS

Heterogeneous GNNs

Type-specific transformations for rooms and wall-surfaces.

Multi-seed validation

Confirm the weighted-BCE + max_iter interaction across independent initializations.

Explainability

Attention-based analysis of individual failure cases.

Thank You

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